ISYE 6740 – Fall 2024
Final Project

Team #: Group 89
Group Members: Andrea Tardif (atardif3@gatech.edu), Jin Dai (jdai79@gatech.edu), Kyla Tracy (ktracy8@gatech.edu)
Project Title: Hospital Bill Predication Using Claims Data

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1 Problem Statement

Overview

Hospital cost prediction using claims data is crucial in healthcare management, insurers, and patients. Since the increased financial burden on healthcare providers and patients, hospital cost prediction has become an essential topic in healthcare analytics. It supports institutions in allocating resources, managing risk, and even making policies. Claims data, consisting of patient demographics, diagnosis codes, procedures, and insurance information, is one of the most valuable sources for making such predictions. Claims data generally includes patient demographic (age, gender, region), medical history (diagnosis codes, treatment plans, prior hospital visits), procedures performed (procedure codes, surgery type, test results), medication prescribed, length of stay, and insurance plan details (copays, deductibles, insurer-paid portion).

Even though claims data can be informative for cost prediction, it requires delicate application. Data quality, case complexity, privacy concerns, nature dynamics can also be analyzed in gauging whether the analysis is applicable. Claims data tends to be noisy and incomplete, each medical procedure and treatment can be highly individualized, whether private information can be used for analysis, and whether nature dynamic should be also considered, such as COVID environment might highly affect the cost for the case. We should thoroughly consider the above questions to utilize claims data. In summary, predicting hospital bills using claims data involves using historical data to model and forecast healthcare costs, helping various stakeholders plan for future expenses and improve operational efficiency. However, the complexity of healthcare data, privacy issues, and evolving medical practices present ongoing challenges in this field.

Literature Review

We explored some major papers that researched claims data. According to **Zhao et al. (2018),** claims data is used for cost prediction and assessing healthcare utilization and outcomes. These longitudinal data cover a broad patient population, making them suitable for prediction. Early studies predicted hospital costs using traditional statistical models, which focused on the linear relationship between the variables and the cost. Obermeyer et al. (**2019)** examined the use of predictive modeling with historical claims data to reduce hospital readmissions and improve transparency in hospital billing. Manning and Mullahy (2001) emphasized the usage of structured claims data (i.e. procedure codes and billing information) for cost prediction. Regression models were used to predict total hospital costs and found that including diagnostic and demographic data improved prediction accuracy. In a study by Zuvekas and Olin (2009), linear regression models were used to estimate healthcare expenditures in the U.S., revealing the statistical meaningfulness of this method for initial cost prediction. Manning and Mullahy (2001) introduced the use of Generalized Linear Model for healthcare cost data, which often shows overdispersion and non-linearity. They suggested that GLMs perform better than standard linear regression in capturing cost variability across different types of patients.

Nowadays, machine learning methods have been introduced to research. As noted by **Obermeyer et al. (2019)**, Random Forest models have been used to predict healthcare costs and outcomes by aggregating the predictions of multiple decision trees. Random Forests are particularly effective in scenarios with large, complex datasets, such as claims data. Wang et al. **(2020)** proved this by applying random forests to predict hospital costs based on claims data. Their model performed better than regression models which indicated that random forests can also identify non-linear relationships between predictor variables like patient demographic and treatment complexity. A study by Zhao et al. (2019) demonstrated Gradient Boosting Machines, known for their superior performance in structured data tasks, such as XGBoost and LightGBM, outperformed traditional regression models in inpatient healthcare cost prediction because they capture the non-linearities in claim data. **Choi et al. (2017)** explored the usage of support vector machines (SVMs) for cost prediction and demonstrated that SVMs are effective in predicting hospitalization costs with a focus on minimizing overfitting through cross-validation. **Zhao et al. (2020)** in their study on cost prediction of emergency room visits found that a hybrid model of decision trees and deep learning performed better than individual modes by predicting emergency room costs with a mean absolute error (MAE) 10% lower than other approaches.

In making successful hospital bill predictions, the transformation of raw claims data into a structured format suitable for machine learning models is important. **Saha et al. (2018)** investigated feature extraction techniques stressing the usage of encoding diagnostic and procedural codes into numerical features. Their findings determined that number of procedures, complexity of diagnosis, and patient's prior medical history were significant predictors of the final hospital bill.

There are many challenges in using claims data to predict hospital costs, i.e. the quality of data (there are inaccuracies in the data such as missing values or incorrect coding), the inherent complexity and variability in patient care, treatment protocols, and billing procedures, and model interpretability. **Liu et al. (2023)** stressed that dealing with missing data and outlier detection was crucial in improving prediction reliability. **Bertsimas et al (2018)** discovered that even sophisticated machine learning models had a difficult time predicting high complexity cases.

The usage of predicting hospital costs is useful in various ways, it can help with efficient resource allocation in hospitals to insurance pricing and patient cost management. **Alberts et al. (2019)** used machine learning to forecast the demand of ICU beds based on historical data. **Xia et al. (2015)** demonstrated that an anomaly detection model could effectively identify fraudulent claims. **Goh and Teo (2020)** showed that predictive modeling helped insurance companies adjust insurance premiums based on hospital expenditure and patient chronic conditions.

As healthcare data becomes increasingly complex, integrating new data sources and advancements in machine learning are expected to improve the accuracy and applicability of hospital cost prediction models further. This will ultimately lead to better resource allocation, cost management, and patient triage in the healthcare sector.

2 Data Source

Our data source for this project is a healthcare dataset from Kaggle (2). This dataset is a synthetic healthcare dataset that was designed to mimic real-world healthcare data. The datasets have 55,500 rows and 15 columns. Each column provides specific information about the patient, their admission, and the healthcare services provided. Below is a brief explanation of each column:

- **Name:** Name of the patient associated with the healthcare record.
- **Age:** The patient's age at the time of admission is expressed in years.
- **Gender:** Indicates the patient's "Male" or "Female."
- **Blood Type:** Patient's blood type (e.g., "A+", "O-", etc.).
- **Medical Condition:** Primary medical condition or diagnosis associated with the patient
- **Date of Admission:** The date the patient was admitted to the healthcare facility.
- **Doctor:** The name of the doctor responsible for the patient's care during their admission.
- **Hospital:** Healthcare facility or hospital where the patient was admitted.
- **Insurance Provider:** Patient's insurance provider (e.g., "Aetna," "Blue Cross," "Cigna," "UnitedHealthcare," and "Medicare.")
- **Billing Amount:** The amount of money billed for the patient's healthcare
- **Room Number:** The room number where the patient was accommodated during their admission.
- **Admission Type:** Specifies the type of admission, which can be "Emergency," "Elective," or "Urgent," reflecting the circumstances of the admission.
- **Discharge Date:** The date on which the patient was discharged from the healthcare facility
- **Medication:** Identifies a medication prescribed or administered to the patient during their admission. Examples include "Aspirin," "Ibuprofen," "Penicillin," "Paracetamol," and "Lipitor."
- **Test Results:** Medical test conducted ("Normal," "Abnormal," or "Inconclusive")

3 Methodology

For our project, we used the healthcare dataset described above to create a predictive model for hospital bills. With the labeled features, we trained a supervised learning methodology to make the prediction model. As shown in Figure 1, we first performed an exploratory analysis of all the attributes in the dataset. Then, we applied the proposed machine learning models to the potential relationship we discovered in previous procedures. Lastly, we evaluated the results of each model and processed predictions using testing datasets.

Exploratory Data Analysis

The original data has 55500 rows of datasets. During the exploratory analysis, we found that the health dataset has no null values and 534 duplicated rows, which had been dropped in the beginning. Then, we checked the potential independent variables and removed the 106 rows that have *Billing Amount* or *Inpatient Days* less than 0. 54860 rows of data are included for further analysis.

Feature Engineering New Variables

Feature engineering was performed to enhance the dataset for analysis. Initially, rows with *Billing Amount* less than or equal to zero were removed to ensure meaningful calculations. Additional features were then derived: *Per Day Amount* was calculated by dividing the *Billing Amount* by *Inpatient Days*, and a logarithmic transformation, *Log Per Day Amount*, was applied to the *Per Day Amount* to normalize its distribution. *Inpatient Days* were calculated as the difference between *Discharge Date and Date of Admission*, plus one to include the day of admission. Furthermore, temporal features such as *Admission Year* and *Admission Month* were extracted from the *Date of Admission*. A total of 5 extra columns are created to the original datasets, they are *Inpatient Days, Per Day Amount, Log Per Day Amount, Admission Year, Admission Month*.

Evenly Distributed Categorical Variables

The dataset comprises an approximately equal split between most categorical data. These categorical variables include G*ender, Blood Type, Medical Condition, Insurance Provider, Admission Type, Medication, Test results*. For example, the G*ender* variable has nearly 50% to 50% distributions between male and female, the *Insurance Provider* has 20% distribution to each categories including Cigna, Medicare, United Healthcare, Blue Cross, Aetna. *Doctor* and *Hospital* variables have numerous categories which can be analyzed further. Below figures show all the categories. *Doctor* has 40341 categories and *Hospital* has 39876 categories.

Next, we prepped and transformed the data into a format suitable for machine learning models. We leveraged label encoding to convert the categorical variables into numeric labels except for *Doctor* and *Hospital*. Then, One-Hot encoding for the gender column was used to transform it into binary columns: *Gender_Female* and *Gender_Male.* Each row has a 1 in one column and a 0 in the other, representing the gender. The same was performed for other categorical columns. Lastly, we removed any invalid rows (i.e., the Billing Amount or Inpatient days were less than 0).

```
Michael Smith      27
John Smith         22
Robert Smith       21
James Smith        20
Michael Johnson    20
                   ..
Shane Tate          1
Christy Parker      1
Larry Miller        1
Chelsea Neal        1
Jeffrey Moore       1
Name: Doctor, Length: 40341, dtype: int64
```

Figure 1: Doctor Category Summary

```
LLC Smith                       44
Ltd Smith                       39
Johnson PLC                     37
Smith Ltd                       37
Smith Group                     36
                                ..
PLC Navarro                      1
PLC Mcintosh                     1
and Hernandez, Hughes Walton     1
Myers-Williams                   1
Moreno Murphy, Griffith and      1
Name: Hospital, Length: 39876, dtype: int64
```

Figure 2: Hospital Category Summary

Numeric Independent Variables

We have *Inpatient Days, Room Number, Admission Years, Admission Month, Age* 5 numeric variables. We plotted a scatter plot and histogram between each numeric variable with *Log Per Day Amount*. We can see that, though they are numeric variables, they can also be seen as categorical. We only see the *Inpatient Days* preserve a potential negative relationship with *Log Per Day Amount*, which can be analyzed further on feature selection.

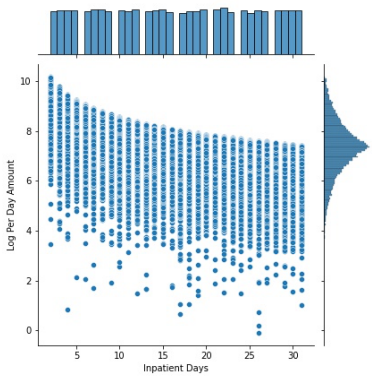


Figure 3: Inpatient Days VS. Log Per Day Amount

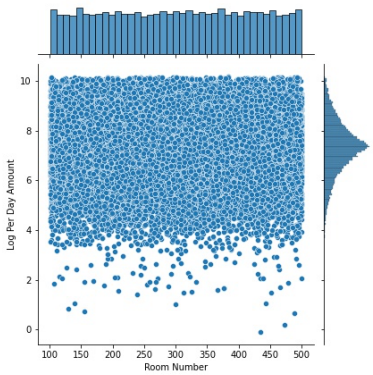


Figure 4: Room Number VS. Log Per Day Amount

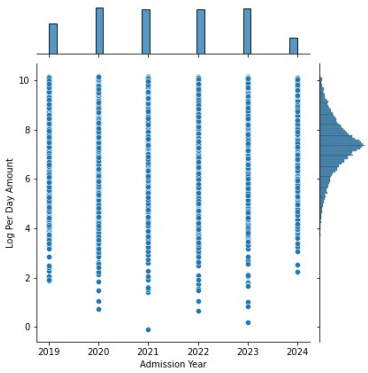


Figure 5: Admission Year VS. Log Per Day Amount

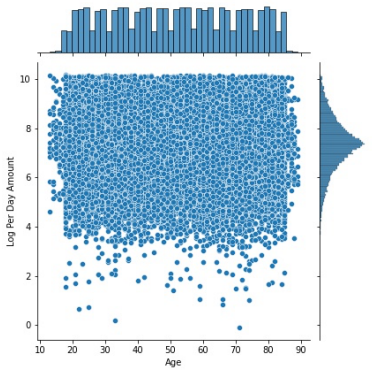


Figure 6: Admission Month VS. Log Per Day Amount

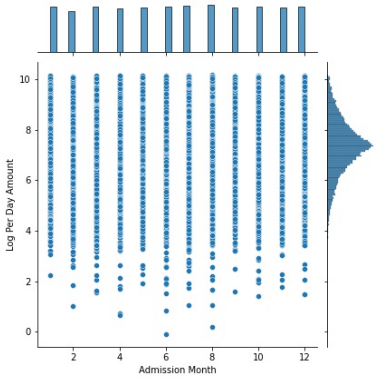


Figure 7: Age VS. Log Per Day Amount

Transformation For Dependent Variable

On the variables we created previously, *Per Day Amount* and *Log Per Day Amount* are transformation for the *Billing Amount*. We plotted the distribution of these variables itself from below figures, we can see the *Billing Amount* is between 0 to 55000, and all values seems to be evenly distributed. Then, we divided *Billing Amount* by *Inpatient Days* to get the *Per Day Amount*. From the *Figure 9*, we can see that the *Per Day Amount* has fewer cases close to 0 and 10000. This also makes more sense for prediction practically, as this is a more normalized variable across the board. We can naturally think about how much would potentially bill every single day per patient, when we have more information for this specific patient. Although, we still see the Distribution for *Per Day Amount* skewed to the right, so we did another log transformation to make the distribution more normal alike. We can see from the histogram of *Log Per Day Amounts* more like a normal distribution, with a mean of 7.30 and a standard deviation of 1.11 from the summary (*Figure 11*).

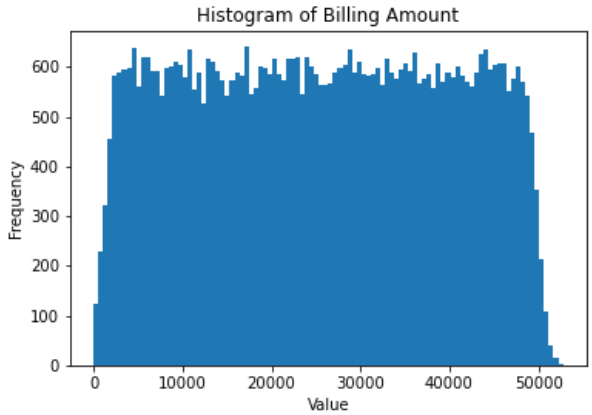


Figure 8: Histogram of Billing amount

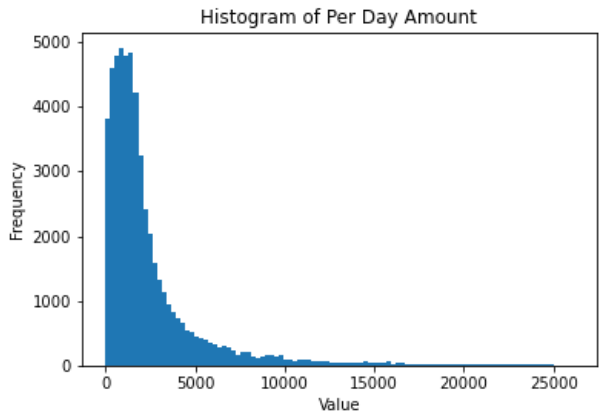


Figure 9: Histogram of Per Day Amount

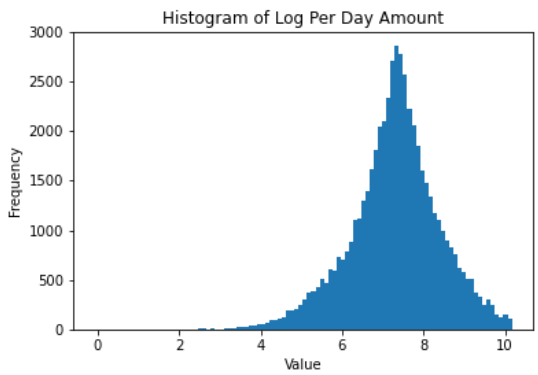


Figure 10: Histogram of Log Per Day Amount

```
count    54860.000000
mean         7.296886
std          1.107542
min         -0.091444
25%          6.697321
50%          7.347310
75%          7.958475
max         10.169918
Name: Log Per Day Amount, dtype: float64
```

Figure 11: Summary for Log Per Day Amount

On top of the histogram, we also plotted the categorical variables to the above potential independent variables, and the figure is included in the appendix. We can see more obvious differences between categories from the plot between categorical variables to *Log Per Day Amount.* We suggest *Log Per Day Amount* as independent variables for future analysis, and this has been double tested by modeling prediction for both *Billing Amount* and *Log Per Day Amount*.

Feature Selection

Mutual Information

We then leveraged mutual information to measure the dependency between features and the *Billing Amount* to identify features that contribute the most predictive power to the model. As shown in *Figure 12*, the top features were the ***Per Day Amount, Hospital,*** and ***Room Number***. These features demonstrated the highest mutual information scores with the Billing Amount, indicating they have the strongest influence on predicting the target.

We performed the same analysis for *Log Per Day Amount* to identify features that contribute the most predictive power. As shown in *Figure 14*, the top features were the ***Per Day Amount, Inpatient Days,*** and ***Hospital***.

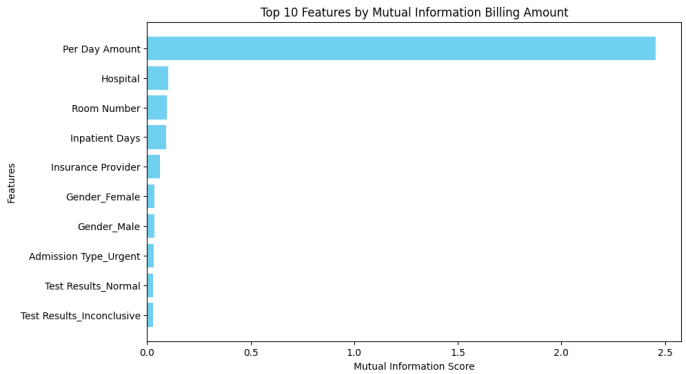


Figure 12: Bar Chart of Mutual Information

	Feature	Billing Amount
0	Per Day Amount	2.456424
1	Hospital	0.101639
2	Room Number	0.096522
3	Inpatient Days	0.092823
4	Insurance Provider	0.061743
5	Gender_Female	0.033836
6	Gender_Male	0.033836
7	Admission Type_Urgent	0.030986
8	Test Results_Normal	0.029913
9	Test Results_Inconclusive	0.029732
10	Admission Type_Emergency	0.029694

Figure 13: Summary of Mutual Information Values

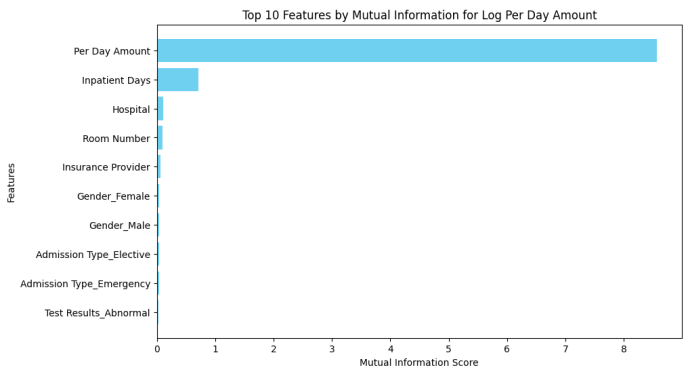


Figure 14: Bar Chart of Mutual Information

	Feature	Log Per Day Amount
0	Per Day Amount	8.567444
1	Inpatient Days	0.712415
2	Hospital	0.102149
3	Room Number	0.100765
4	Insurance Provider	0.061740
5	Gender_Female	0.036009
6	Gender_Male	0.036002
7	Admission Type_Elective	0.033259
8	Admission Type_Emergency	0.032741
9	Test Results_Abnormal	0.029422
10	Admission Type_Urgent	0.028611

Figure 15: Summary of Mutual Information Values

Chi-Square Statistic

We also performed a chi-square analysis to test the categorical feature's independence from the target variable. As shown in *Figure 14*, the two categorical variables with a p-value less than 0.05 were ***Medical Condition*** and ***Hospital***.

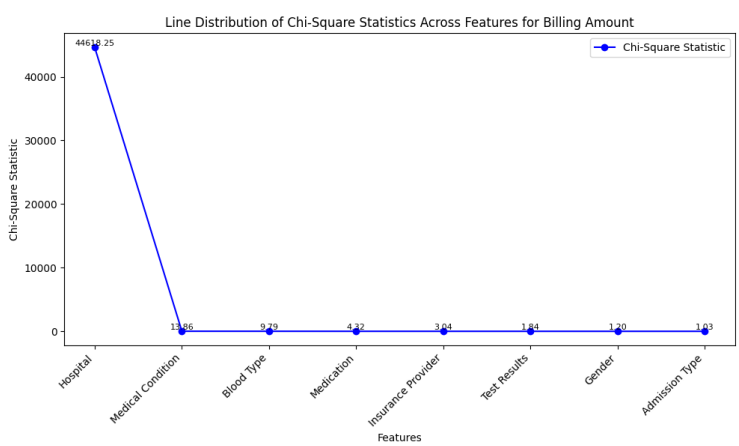


Figure 14: Line Distribution of Chi-Square Test for Billing Amount

	chi2_stat	p_value
Medical Condition	13.859433	1.652740e-02
Hospital	44618.248099	3.770287e-59
Gender	1.199352	2.734512e-01
Blood Type	9.789184	2.008383e-01
Admission Type	1.026659	5.984996e-01
Medication	4.324905	3.638169e-01
Test Results	1.840273	3.984647e-01
Insurance Provider	3.044129	5.504679e-01

Figure 15: Summary of Chi-Square Test for Billing Amount

Again, we performed the same chi-square analysis to test the categorical feature's independence from the target variable of *Log Amount Per Day.* As shown in *Figure 16*, the two categorical variables with a p-value less than 0.05 were ***Medical Condition*** and ***Hospital***.

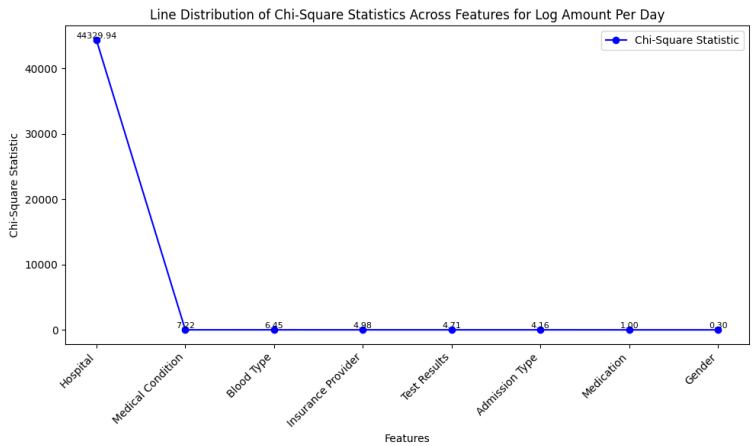


Figure 16: Line Distribution of Chi-Square Test for Log Per Day Amount

	chi2_stat	p_value
Medical Condition	7.218017	2.049245e-01
Hospital	44329.940064	1.194680e-52
Gender	0.304068	5.813430e-01
Blood Type	6.446444	4.886906e-01
Admission Type	4.159462	1.249638e-01
Medication	0.998045	9.100924e-01
Test Results	4.713265	9.473872e-02
Insurance Provider	4.978812	2.894784e-01

Figure 17: Summary of Chi-Square Test for Log Per Day Amount

4 Model Selection

Model selection was a key step in this project. We reviewed several supervised machine learning models discussed in class and selected the ones that aligned best with our objectives. For regression-based models, we chose linear regression and lasso regression. Linear regression was selected because of its simplicity and its utility as a baseline model, operating under the assumption that hospital billing prices might have a linear relationship with some of the features. Lasso regression, on the other hand, was chosen for its ability to perform automatic feature selection, which helps improve accuracy while allowing for easy tuning to balance model complexity and performance.

For tree-based models, we decided to evaluate Random Forest, K-Nearest Neighbors (KNN), and Gradient Boosting. Random Forest was chosen for its ability to handle non-linear relationships while averaging across multiple trees, which increases robustness to noise and reduces the risk of overfitting. K-Nearest Neighbors was included due to its ability to capture non-linear patterns and adapt to high-dimensional data with minimal feature selection. Gradient Boosting was selected because of its sequential learning approach, which often enables it to outperform random forest by capturing more nuanced interactions between features.

Lastly, for neural network-based models, we tested MLP Regression and MLP Classification. MLP Regression, a feedforward neural network, was chosen for its capacity to handle large datasets and its ability to model non-linear relationships effectively. Based on the performance of the MLP Regressor, we decided to explore MLP Classification to evaluate how well it could classify prices into ranges rather than predicting exact values.

5 Model Evaluation

After tuning and adding the selected features to each model, we compared the predicted cost to the actual costs. The health dataset was split into training and testing subsets using an 80-20 split ratio, ensuring that 80% of the data was allocated for training and 20% for testing. The error metrics for the models are below:

Model Error Metrics	MSPE		RSME		Accuracy		R^2 Error
	Test	Train	Test	Train	Test	Train	
Linear Regression	-324089	-328478	13075	13117	0.489	0.487	0.15
Lasso Regression	-323816	-328650	13069	13120	0.4893	0.4873	0.15
Random Forest	97.4	99.7	36.9	13.5	0.998	0.9994	0.99
K-Nearest Neighbors	-174141	-168522	9585	9396	0.63	0.632	0.54
Gradient Boosting	-5920	-5578	1775	1731	0.93	0.933	0.984
MLP Regression	-4755	-3987	1564	1468	.94	.943	.988
MLP Classifier	96.2	95.7	1.05	1.32	0.98	0.982	0.99

Table 1: Prediction Model for Billing Amount Error Metrics

Model Error Metrics	MSE		RSME		Accuracy		R^2 Error
	Test	Train	Test	Train	Test	Train	
Linear Regression	.76	.768	.872	.876	.88	.88	.38
Lasso Regression	.747	.776	.864	.881	.882	.879	.38
Random Forest	.709	.714	.842	.845	.8845	.8841	.42
K-Nearest Neighbors	.71	.714	.843	.845	.885	.884	.42
Gradient Boosting	.71	.706	.843	.84	.8845	.8848	.42
MLP Regression	.75	.658	.87	.81	.882	.89	.39
MLP Classifier	1.63	1.64	1.277	1.28	.642	.639	-.01

Table 2: Prediction Model for Log Per Day Amount Error Metrics

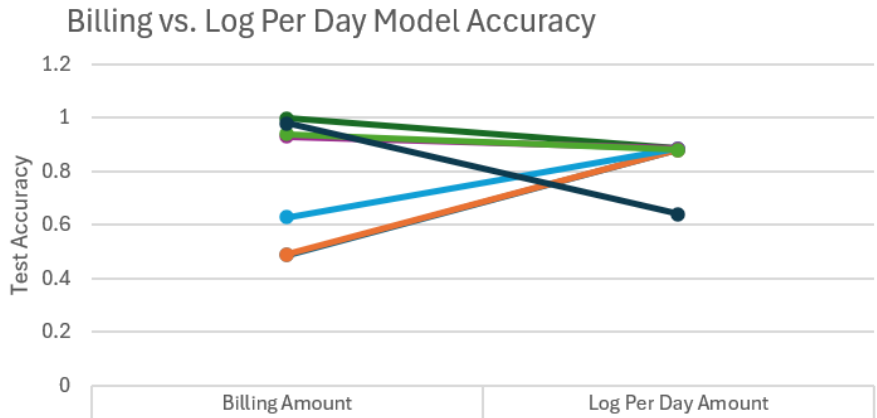
Below are the best features for each model based on their respective selection criteria and performance metrics and the time to run each model:

Target Variable	Model	Time to Run	Features Used
Billing Amount	Linear Regression	0.6 seconds	Medication_Ibuprofen, Medication_Lipitor, Medication_Penicillin, Medication_Aspirin, Medication_Paracetamol, Medical Condition_Hypertension, Condition_Asthma, Medical Condition_Arthritis, Medical Condition_Cancer, Medical Condition_Obesity, Medical Condition_Diabetes, Test Results_Abnorma, Test Results_Normal, Test

			Results_Inconclusive, Blood Type_A+, Blood Type_O+, Blood Type_AB-, Blood Type_B+, Blood Type_B-, Blood Type_A-, Blood Type_AB+, Blood Type_O-, Admission Type_Urgent, Admission Type_Emergency, Admission Type_Elective, Per Day Amount, Inpatient Days
Billing Amount	Lasso Regression	0 seconds	Per Day Amount, Inpatient Days
Billing Amount	Random Forest	0.6 seconds	Per Day Amount, Inpatient Days
Billing Amount	KNN	0.4 seconds	Per Day Amount, Hospital, Inpatient Days
Billing Amount	Gradient Boost	13.6 seconds	Per Day Amount, Inpatient Days
Billing Amount	MLP Regression	3 minutes 2.3 seconds	Per Day Amount, Inpatient Days, Addmision Type_Urgent
Billing Amount	MLP Classifier	18.9 seconds	Per Day Amount, Inpatient Days, Hospital, Age, Admission Type_Elective, Gender
Log Per Day Amount	Linear Regression	0 seconds	All features except Age, Room Number, Hospital, and Insurance
Log Per Day Amount	Lasso Regression	0 seconds	Only Inpatient Days (0.01% difference between all features and only Inpatient Days)
Log Per Day Amount	Random Forest	0 seconds	Only Inpatient Days (minimal difference between all features and only Inpatient Days)
Log Per Day Amount	KNN	0.7 seconds	Inpatient Days only (1000 neighbors)
Log Per Day Amount	Gradient Boost	18 seconds	Inpatient Days, Hospital, Room Number, Insurance Provider
Log Per Day Amount	MLP Regression	51 seconds	Inpatient Days, Admission Type_Urgent, Gender_Male, Admission Type_Emergency, Gender_Female, Admission Type_Elective, Test Results_Abnormal, Test Results_Normal
Log Per Day Amount	MLP Classifier	19.2 seconds	Inpatient Days, Admission Type_Urgent, Gender_Male, Admission Type_Emergency, Gender_Female, Admission Type_Elective, Test Results_Abnormal, Test Results_Normal

Table 3: Features Used and Time per run for each Prediction Model

The MLP regression model and Random Forest were the best in estimating the log of the daily average, while the MLP classification model performed the worst. The Random Forest outperformed both the MLP regressor and MLP classification models, performing the best in estimating the billing amount using the daily average as a feature.



	Billing Amount	Log Per Day Amount
Linear Regression	0.489	0.88
Lasso Regression	0.4893	0.882
Random Forest	0.998	0.885
K-Nearest Neighbors	0.63	0.885
Gradient Boosting	0.93	0.8845
MLP Regression	0.94	0.882
MLP Classifier	0.98	0.642

Figure 18: Graph of Model's Test Accuracy for Billing Amount and Log Per Day Amount

6 Conclusion

We found that the Random Forest and MLP Regression models consistently performed the best for predicting both the billing amount and the log of the daily average. Although the MLP Classification model performed well for billing amount prediction, it did not perform as well as other models when it came to predicting the log per day amount.

Some key findings from our analysis included the importance of transforming dependent variables to improve model performance. The regression models benefited significantly from such transformations, yielding better error metrics compared to models that relied solely on raw data. Furthermore, features such as Inpatient Days, Hospital, and Insurance Provider were pivotal for most models. We also found that we initially omitted variables based on judgment, such as Room Number, but while testing, we found it to be important for most models.

To conclude, this project highlights the effectiveness of regression-based models and the importance of thoughtful feature selection and data transformation. While MLP Regression performed the best, its high computational cost shows the need to balance accuracy and efficiency for future projects.

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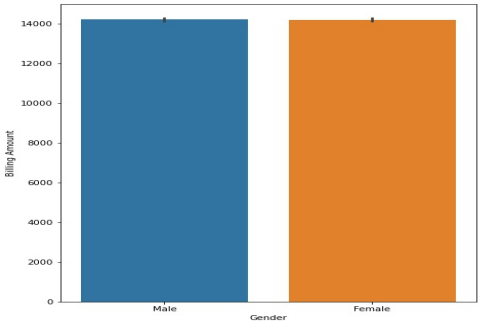
Appendix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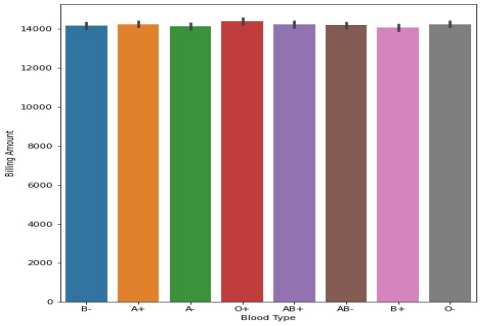
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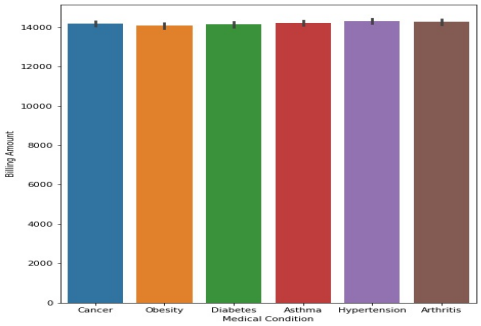
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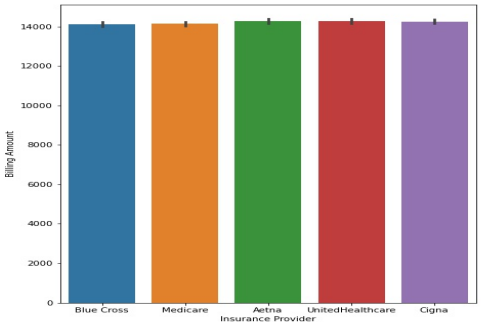
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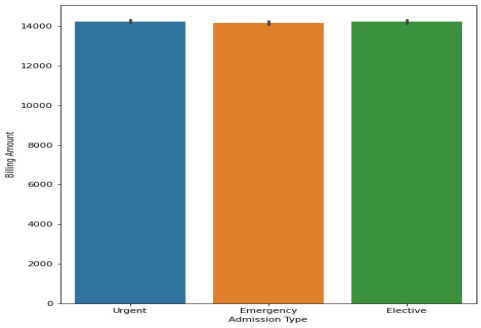
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Emergency
Elective
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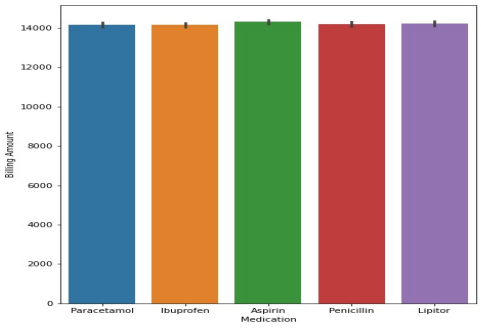
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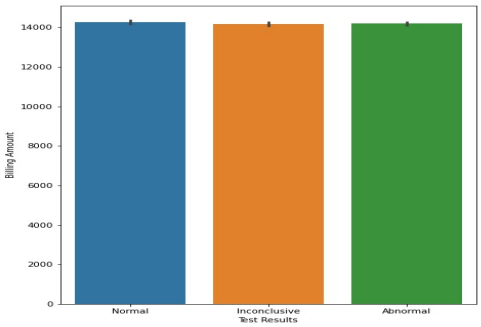
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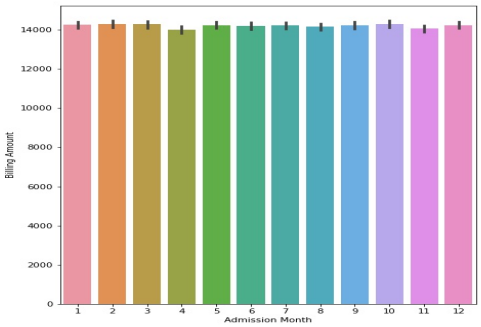
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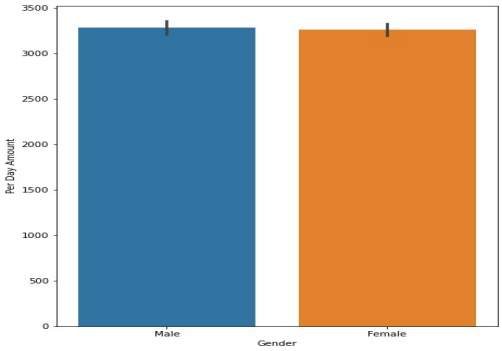
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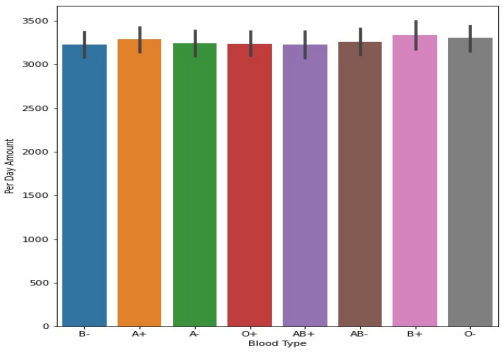
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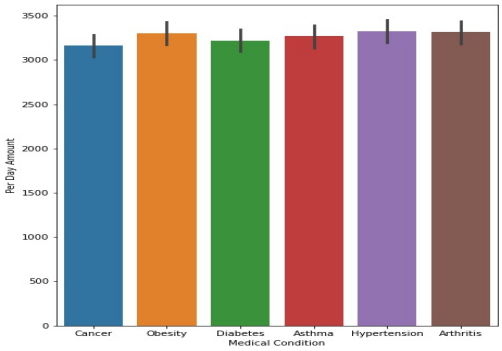
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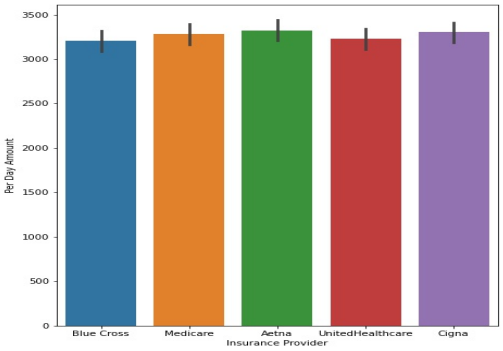
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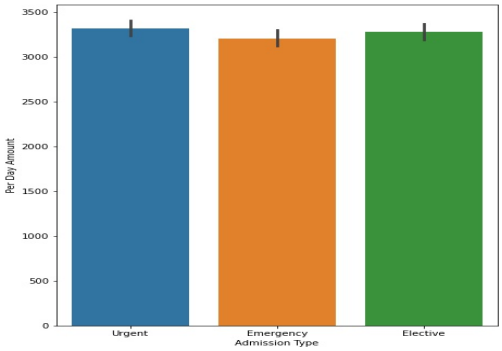
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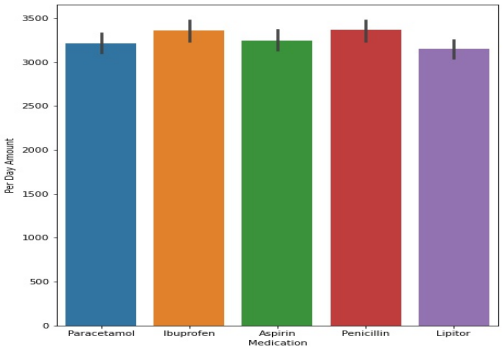
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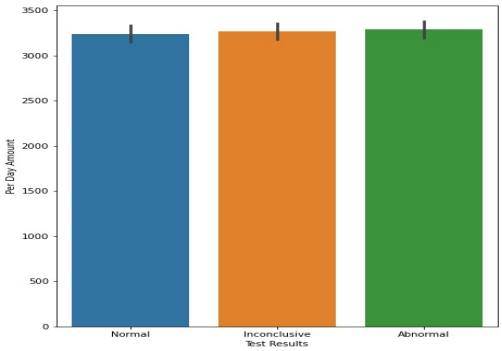
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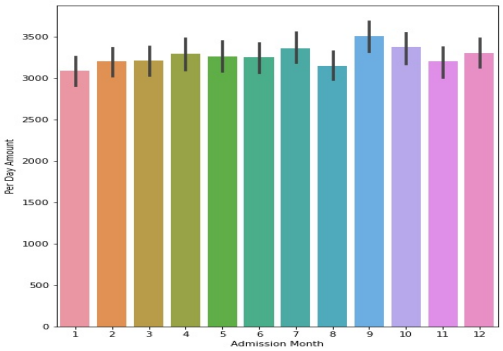
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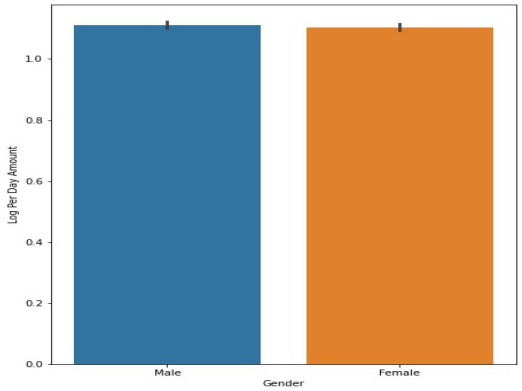
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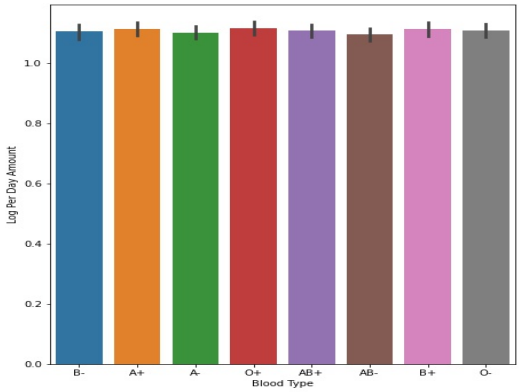
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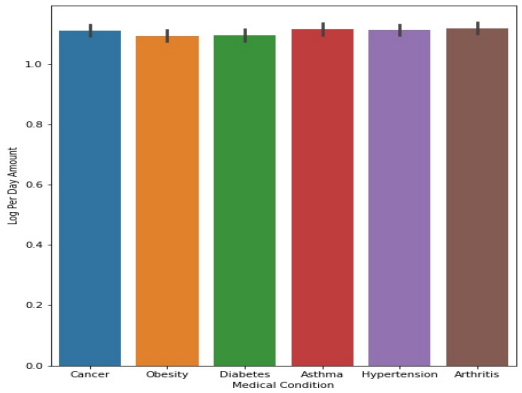
Blood Type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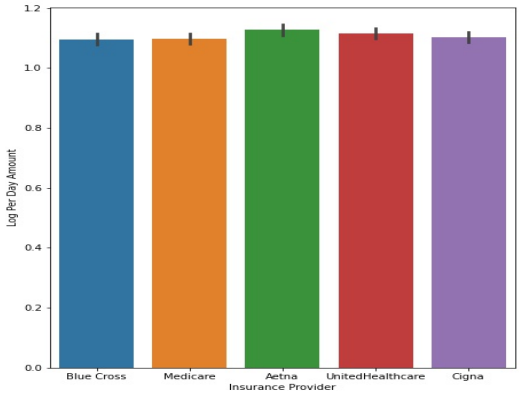
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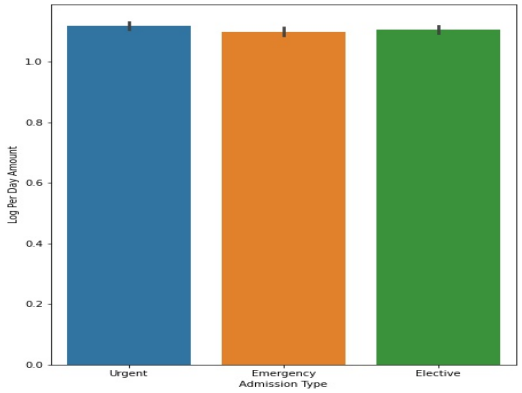
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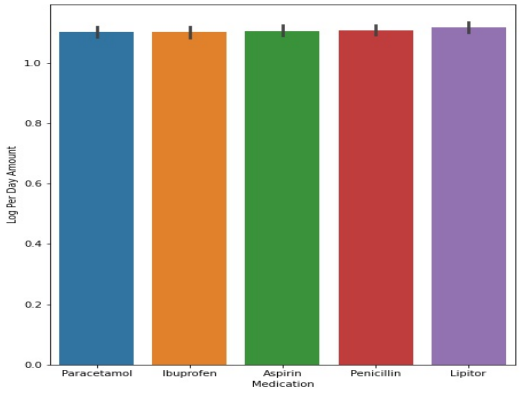
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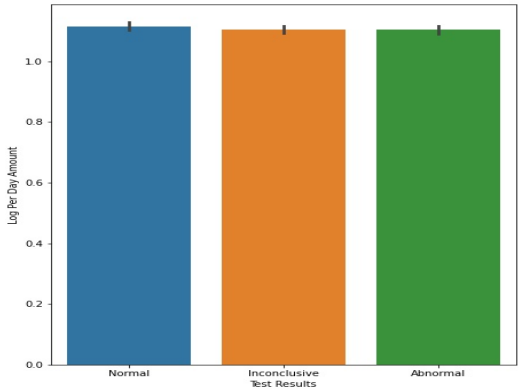
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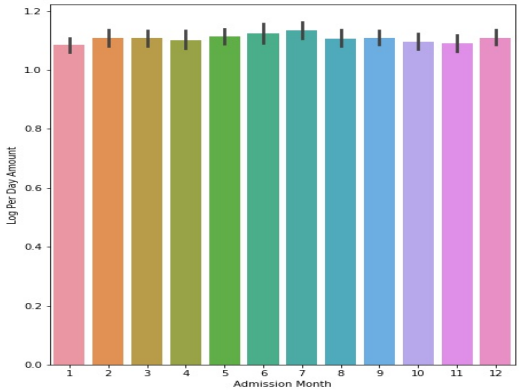
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Admission Month